

| Question |     |       | Marking details |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |                                            |     |       |       | Prac |
|----------|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------|-----|-------|-------|------|
|          |     |       |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2                                        | AO3 | Total | Maths |      |
| 4        | (a) | (i)   |                 | Test wire and reference wire made from the same material or identical (1) Incorrect to refer to common support<br>Temperature change will have the same effect on both wires (1)<br>Don't accept reference to being at the same temperature                                                                                                                                                                                                                                                                     |                 |                                            | 2   | 2     |       | 2    |
|          |     | (ii)  |                 | Extension / point beyond which the wire will not return to its original length / permanently deformed (1)<br>Don't accept reference to Hooke's law or limit of proportionality<br><u>Removing load</u> (or equivalent) and observe whether or not wire returns to original length <b>or</b> when the load is removed the extension values are the same (1)                                                                                                                                                      | 1               |                                            | 1   | 2     |       | 1    |
|          |     | (iii) | I               | Improve accuracy by reducing [fractional] uncertainty <b>or</b> to provide a measurable extension / longer wire <b>or</b> produces greater extension / more accurate extension<br>Don't accept reduce uncertainty only or a wider range of results                                                                                                                                                                                                                                                              |                 |                                            | 1   | 1     |       | 1    |
|          |     |       | II              | Improve accuracy (or reduce uncertainty) in cross-sectional area / to obtain a mean value for diameter / check for uniformity                                                                                                                                                                                                                                                                                                                                                                                   |                 |                                            | 1   | 1     |       | 1    |
|          | (b) | (i)   |                 | $A = 8.04 \times 10^{-8} \text{ [m}^2\text{]} / 0.08 \text{ mm}^2$ (1) (or by implication)<br><br>Load extension combination e.g. 28 N, $4.8 \times 10^{-3} \text{ m}$ (1) (or by implication) e.g. 5833.3<br>Substitution into $E = \frac{Fl}{Ae}$ e.g. $\frac{28 \times 2.4}{8.04 \times 10^{-8} \times 4.8 \times 10^{-3}}$ (1) ( <b>ecf</b> on A and load extension combination)<br>$E = 1.74 \times 10^{11} \text{ N m}^{-2}$ or Pa or sensible alternative (1) <b>unit mark</b><br>Accept 2 or 3 sig figs | 1               | 1<br><br><br><br><br><br><br><br><br><br>1 |     | 4     | 4     |      |